Temporal Analysis and Forecasting of Wildlife Observation Data Using Time Series Methods

Vinay P Ramesh, Darsh Veer Singh, Ahilnandan Kabilan
International Institute of Information Technology Bangalore
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Emails: Vinay.Ramesh@iiitb.ac.in, Darshveer.Singh@iiitb.ac.in, Ahilnandan.Kabilan@iiitb.ac.in

***Abstract*—This paper presents a comprehensive temporal analysis of the India Convention on Migratory Species (CMS) wildlife observation dataset. Because the dataset is event-driven rather than a native time series, a preprocessing pipeline was developed to aggregate individual sighting records into a structured monthly observation-count series spanning May 2019 to December 2025 (80 months, 249 records, 52 species across three taxonomic groups). Exploratory analysis revealed extreme temporal imbalance, a pronounced annual seasonal pattern aligned with the winter migration cycle, and a structural spike of 116 observations concentrated in January 2025. Additive and multiplicative seasonal decomposition yielded a trend strength of $F_T = 0.29$ and a seasonality strength of $F_S = 0.20$, indicating weak-to-moderate components heavily overshadowed by residual noise. Autocorrelation analysis (ACF and PACF on raw, log-transformed, and first-differenced variants) revealed significant lag-1 and lag-12 correlations, consistent with short-term AR(1) dynamics and annual periodicity. A three-tier stationarity framework was applied: strict stationarity assessed via Kolmogorov–Smirnov and Levene tests; second-order stationarity via Augmented Dickey–Fuller (ADF), KPSS, and Ljung–Box tests; and intrinsic stationarity via empirical variogram analysis. The raw and log-transformed series failed all frameworks; the first-differenced series passed ADF and KPSS but retained heteroscedastic variance owing to the dominant 2025 structural spike. For forecasting, a SARIMA model (order selected via AIC grid search as $(1,1,1)(0,1,1)_{12}$) and a seasonal linear regression model were both trained on 68 months and validated against a strict 12-month holdout. Both models produced negative R^2 scores (-0.27 and -0.30 respectively), reflecting an out-of-distribution regime change in the test period rather than model misspecification. Directions for future improvement are discussed.**

I. Introduction

Wildlife monitoring datasets compiled under international conservation frameworks such as the Convention on Migratory Species (CMS) are increasingly being analysed for temporal trends that may signal ecological change. Unlike purpose-built environmental sensor streams, these datasets are inherently event-driven: each record represents a discrete sighting logged at an irregular time by a variable number of observers, subject to reporting bias and inconsistent collection effort. Direct application of classical time series methods therefore requires principled preprocessing, and the results must be interpreted carefully in light of those data characteristics.

This work addresses the full temporal analysis pipeline for the India CMS dataset. The primary contributions are:

- A preprocessing pipeline that converts raw event records into a regular monthly observation-count series.
- Seasonal decomposition (additive and multiplicative) with quantitative trend and seasonality strength metrics.
- Rigorous ACF/PACF analysis across three series variants.
- A three-tier stationarity framework covering strict, second-order (weak), and intrinsic stationarity.
- Comparative forecasting using SARIMA (grid-searched on AIC) and seasonal linear regression, validated on a 12-month holdout.

Three assignment questions are addressed throughout: (i) what dataset characteristics make it amenable to temporal analysis; (ii) whether the chosen methods worked and why; and (iii) what further analyses could deepen understanding.

II. Dataset Description and Pre-processing

The dataset used in this study consists of wildlife observation records of migratory species distributed across India. Each record represents a discrete sighting event and is associated with temporal attributes including species name, taxonomic group, observation type, event date, migration season, and IUCN conservation status.

A. Dataset Characteristics

The dataset exhibits several characteristics that make it suitable for temporal statistical analysis:

- Precise ISO-formatted timestamps enabling unambiguous monthly aggregation across a multi-year window.
- Coverage of 52 species across three taxonomic groups (Aves, Mammalia, and Reptilia), spanning (largely) May 2019 to December 2025.
- The migratory nature of the recorded species creates a prior expectation of annual periodicity, as CMS species sightings peak during the Indian winter migration season (October–February).
- The monthly count structure ($Y_t \in \mathbb{Z}_{\geq 0}$) is compatible with ARIMA-family models after log transformation.

However, the dataset also presents significant challenges: 68.75% of months have zero counts, observations are concentrated in a single month (January 2025), and the series almost certainly confounds genuine ecological trends with changes in data collection effort. Figure 1 provides a clear visual confirmation of this extreme temporal sparsity and concentration.

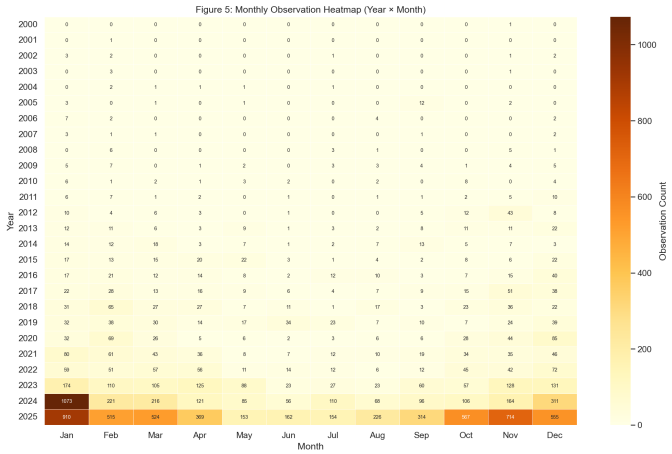

Fig. 1. Heatmap of wildlife observation counts by year and month. This visualization highlights the extreme temporal sparsity of the dataset (dominated by zero-count months across the early years) and the anomalous concentration of records in January 2025.

TABLE I
MONTHLY OBSERVATION COUNT SERIES – SUMMARY STATISTICS

Property	**Value**
Series length (months)	80
Total observation records	249
Unique species	52
Taxonomic groups	3 (Aves, Mammalia, Reptilia)
Mean monthly count	3.11
Standard deviation	13.29
Min / Max	0 / 116
Non-zero months	25 of 80 (31.3%)
Variance – first half	0.12
Variance – second half	334.78
Skewness	7.72
Excess kurtosis	62.13

Table I summarises the key properties of the constructed monthly time series. The series is highly right-skewed (skewness $= 7.72$, excess kurtosis $= 62.13$), dominated by the structural spike of 116 observations in January 2025. The variance in the first half ($\sigma^2 = 0.12$) is more than 2800 times smaller than the second half ($\sigma^2 = 334.8$), flagging severe heteroscedasticity.

B. Pre-processing Steps

Pre-processing was carried out in Python to ensure data consistency and analytical readiness. The key steps were:

- Removal of spatial and administrative columns (`latitude`, `longitude`, `country`, `state_province`, `occurrence_id`).
- Parsing of `event_date` with `pandas.to_datetime(errors='coerce')`, dropping rows with invalid dates.
- Filtering to records from 2018 onwards, yielding 249 records from an original 12,374 raw entries.
- Aggregation into a regular monthly count series using `pandas.resample('MS')`, with zero-filling for months containing no observations.

The distribution of observations across years and months is shown in Figure 2. The year-wise panel confirms that 2025 dominates (189 of 249 records), while the month-wise panel shows January as the peak month, consistent with the Indian winter migration season.

Individual events were aggregated into a regular monthly count:

$$Y_t = \left|\{i : \text{event}_i \in \text{month } t\}\right| \tag{1}$$

A log-plus-one transformation was applied to stabilise variance:

$$Y_t^{(\log)} = \log(1 + Y_t) \tag{2}$$

For the multiplicative decomposition, zeros were replaced with $\epsilon = 0.1$ to avoid division by zero.

III. SEASONAL DECOMPOSITION

A. Time Series and Rolling Statistics

Figure 3 shows the raw monthly count series overlaid with 3-month and 12-month rolling means (top panel) and the corresponding rolling standard deviation (bottom panel). Both the rolling mean and rolling standard deviation are near zero from 2019 to 2022, then rise sharply in 2025, visually confirming non-constant mean and variance. The series is therefore *not* stationary in its raw form.

B. Additive and Multiplicative Models

Classical seasonal decomposition [1] expresses the series as either:

$$Y_t = T_t + S_t + R_t \tag{3}$$

$$Y_t = T_t \cdot S_t \cdot R_t \tag{4}$$

where T_t is the trend component, S_t is the seasonal component with assumed period $m = 12$, and R_t is the residual. Decomposition was performed using `statsmodels.tsa.seasonal.seasonal_decompose` with `period=12` and `extrapolate_trend='freq'`.

Figure 4 shows the additive decomposition. The trend component rises gradually from 2020 then accelerates sharply in late 2024, peaking in January 2025. The seasonal component reveals a repeating annual wave with peaks in winter months. The residual is dominated by the January 2025 spike, which the trend and seasonal components alone cannot capture.

Figure 5 shows the multiplicative decomposition. The seasonal factors are larger in magnitude during recent years when the trend is higher, consistent with variance scaling with the level of the series. The multiplicative residual is more homogeneous than in the additive case during the 2025 peak, suggesting a better fit for that period.

C. Trend and Seasonality Strength

Following the variance-ratio framework of [9], the strengths of trend and seasonality components are:

$$F_T = \max\left(0,\ 1 - \frac{\text{Var}(R_t)}{\text{Var}(T_t + R_t)}\right) \tag{5}$$

$$F_S = \max\left(0,\ 1 - \frac{\text{Var}(R_t)}{\text{Var}(S_t + R_t)}\right) \tag{6}$$

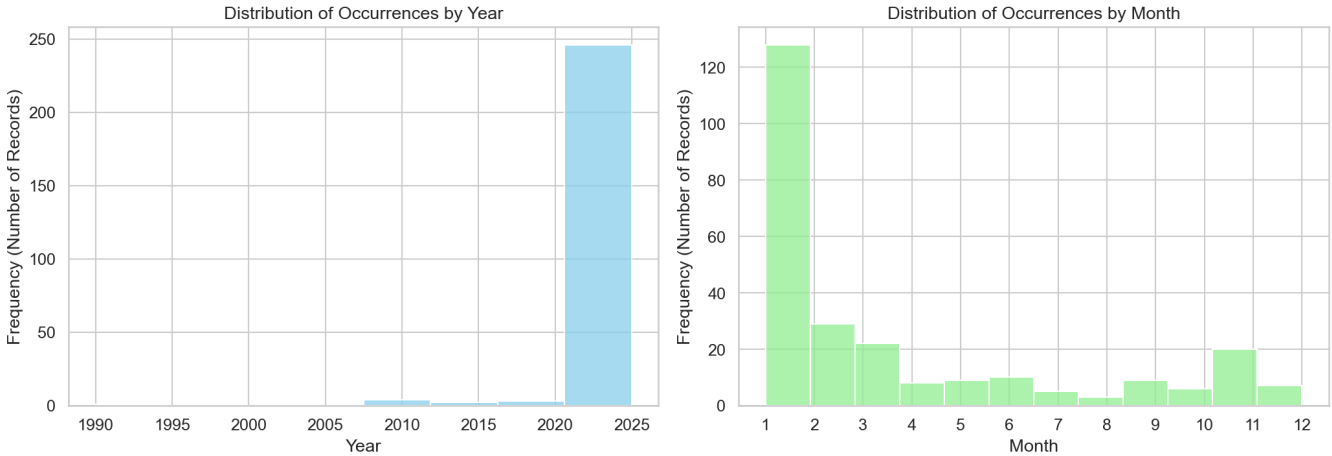

Fig. 2. Year-wise (left) and month-wise (right) distribution of wildlife observation records. Observations are heavily concentrated in 2025 and in the winter months (January–March), consistent with the Indian migratory season.

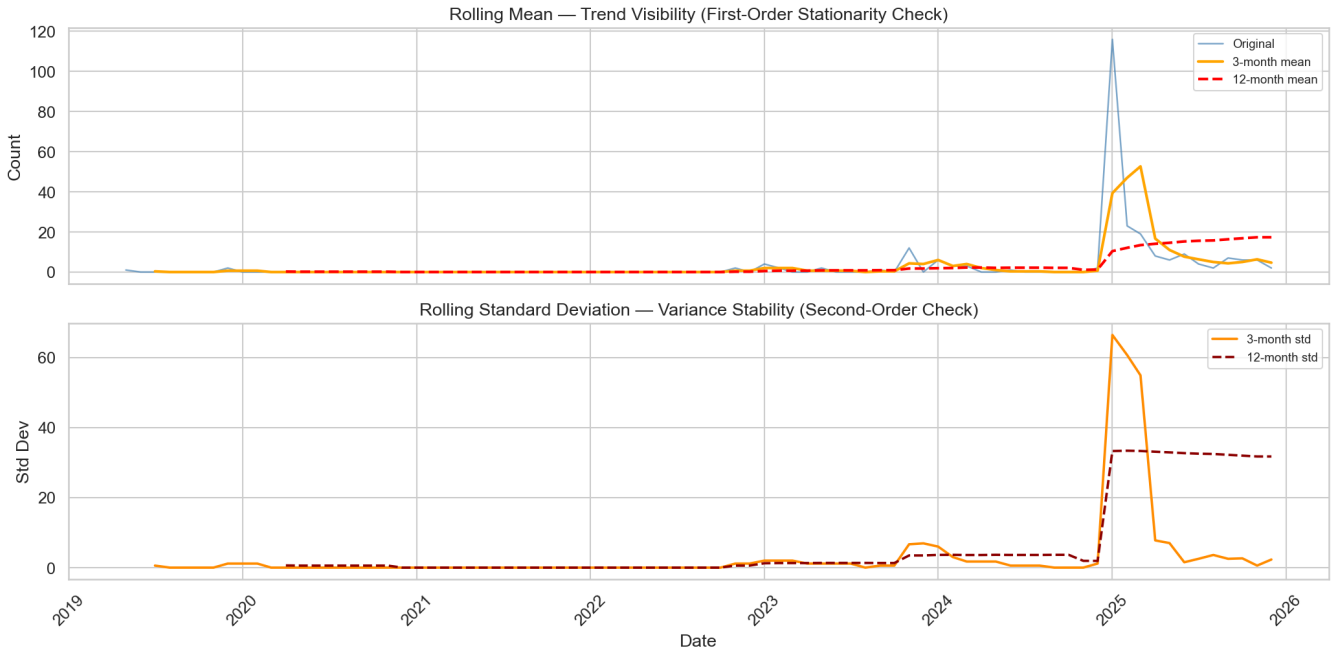

Fig. 3. Top: raw monthly observation count with 3-month and 12-month rolling means. Bottom: rolling standard deviation. Both statistics are near zero before 2023 and spike sharply in 2025, confirming non-stationary mean and variance in the raw series.

The additive decomposition yields $F_T = 0.2949$ and $F_S = 0.1999$. Both values are weak to moderate, reflecting that the residual variance is large relative to the explained components. This is a direct consequence of the dominant January 2025 spike, which inflates residual variance and suppresses the apparent strength of the underlying pattern. While trend and seasonality are present and ecologically real, they are not strong enough to drive reliable long-range forecasts.

IV. Autocorrelation Analysis

A. ACF and PACF

Autocorrelation analysis was performed on three series variants: raw Y_t, log-transformed $Y_t^{(\log)}$, and first-differenced $\Delta Y_t = Y_t - Y_{t-1}$. The Autocorrelation Function at lag k is:

$$\hat{\rho}(k) = \frac{\sum_{t=k+1}^{n} (Y_t - \bar{Y})(Y_{t-k} - \bar{Y})}{\sum_{t=1}^{n} (Y_t - \bar{Y})^2} \tag{7}$$

The Partial Autocorrelation Function (PACF) removes the indirect influence of intermediate lags via the Yule–Walker equations solved by the Levinson–Durbin recursion [2]. The 95% confidence bounds are $\pm 1.96/\sqrt{n}$.

Figure 6 shows all six ACF and PACF panels.

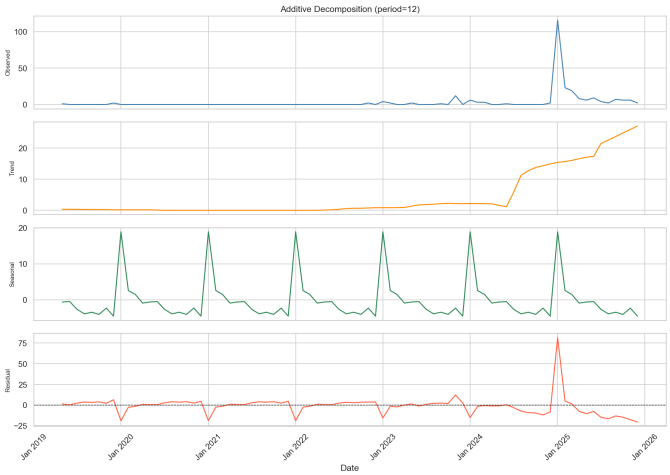

Fig. 4. Additive seasonal decomposition (period=12). From top to bottom: observed series, trend component, seasonal component, and residual. The residual is dominated by the January 2025 structural spike, indicating an anomalous event beyond the model's explanatory power.

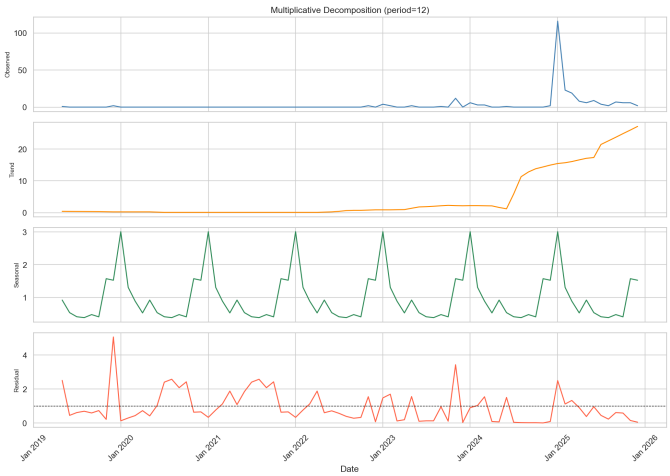

Fig. 5. Multiplicative seasonal decomposition (period=12). Seasonal factors scale with the trend level, producing a more homogeneous residual than the additive model during the 2025 peak period.

Raw series. The ACF shows a slow, persistent decay with a significant positive spike at lag 1 and another at lag 12, confirming short-term autocorrelation and annual periodicity. The PACF shows a single dominant spike at lag 1 followed by values mostly within the confidence band – the hallmark signature of an AR(1) process.

Log-transformed series. The ACF pattern is qualitatively similar but with attenuated magnitude. Lag 1 and lag 12 remain significant. The log transformation compresses the spike without changing the qualitative autocorrelation structure.

First-differenced series. The ACF shows one significant negative spike at lag 1, characteristic of over-differencing, with all remaining lags falling inside the confidence band. This near-white-noise pattern confirms that $d = 1$ differencing is sufficient to remove temporal dependence in the mean, supporting $I(1)$ in the SARIMA model.

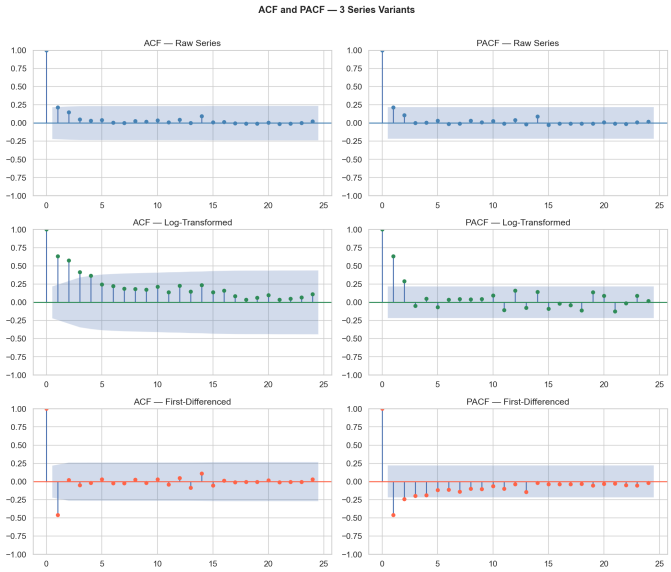

Fig. 6. ACF (left column) and PACF (right column) for three series variants: raw count (top), log-transformed (middle), and first-differenced (bottom). Shaded regions are 95% confidence bounds. Significant lag-1 and lag-12 spikes in the raw and log series indicate AR(1) dynamics and annual periodicity; the differenced series is near white noise.

B. Lag Scatter Plots

Figure 7 shows scatter plots of Y_t versus Y_{t-k} for lags $k \in \{1, 3, 6, 12\}$ with fitted regression lines and Pearson correlation coefficients. Lag 1 shows the strongest association (consistent with AR(1) structure). The January 2025 outlier is visible as a high-value point far from the near-zero cluster in all panels.

V. Stationarity Analysis

A. Strict Stationarity

Definition. $\{Y_t\}$ is strictly stationary if its joint distribution is invariant to arbitrary time shifts:

$$F(y_{t_1}, \ldots, y_{t_k}) = F(y_{t_1+\tau}, \ldots, y_{t_k+\tau}) \quad \forall\, k, \tau, t_1, \ldots, t_k \tag{8}$$

KS two-sample test. The series was split at its midpoint (first half: May 2019–December 2022; second half: January 2023–December 2025) and empirical CDFs compared:

$$D_{n,m} = \sup_x \left| F_n(x) - G_m(x) \right| \tag{9}$$

The raw series gives $p = 0.000\,02$, strongly rejecting identical distributions. The first half contains only values of 0, 1, or 2; the second half contains values up to 116. The first-differenced series also fails ($p = 0.004$) because the 2025 spike creates large negative increments absent from the training window. *Strict stationarity fails for all series variants.*

Levene's test [8]. Splitting the series into four equal windows yields $p = 0.043$ for the raw series, rejecting equal variance across time windows.

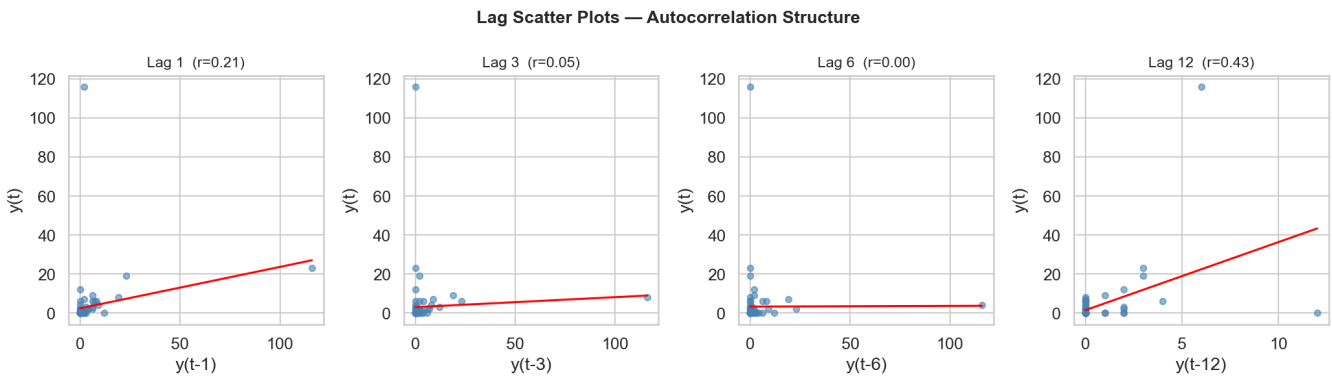

Fig. 7. Lag scatter plots of Y_t versus Y_{t-k} for lags $k \in \{1, 3, 6, 12\}$ with fitted regression lines and Pearson r values. The January 2025 outlier is visible as a high-x or high-y point in all panels, dominating the correlation at longer lags.

B. Second-Order (Weak) Stationarity

Definition. $\{Y_t\}$ is second-order stationary if: (i) $\mathbb{E}[Y_t] = \mu$ (constant mean); (ii) $\mathrm{Var}(Y_t) = \sigma^2 < \infty$ (constant variance); (iii) $\mathrm{Cov}(Y_t, Y_{t+k}) = \gamma(k)$ (autocovariance depends only on lag k).

Mean stability (Welch's t-test). The raw series gives $p = 0.045$ (mean of half 1: 0.075; mean of half 2: 6.15), confirming the mean is not constant. The first-differenced series gives $p = 0.984$, confirming a stable mean after differencing.

Variance stability (Levene's test). Even after first differencing, variance of half 1 is 0.23 versus 556.4 for half 2 ($p = 0.044$), indicating persistent heteroscedasticity due to the 2025 spike.

ADF test [3]. The ADF regression:

$$\Delta Y_t = \alpha + \phi Y_{t-1} + \sum_{j=1}^{p} \delta_j \Delta Y_{t-j} + \varepsilon_t \tag{10}$$

tests H_0: unit root. For the raw series, ADF $p = 0.000$ (rejects unit root). For the log-transformed series, ADF $p = 0.087$ (fails). For the first-differenced series, ADF $p = 0.000$ (passes decisively).

KPSS test [4]. H_0: stationarity.

$$\eta_\mu = \frac{1}{n^2 \hat{\sigma}_\ell^2} \sum_{t=1}^{n} S_t^2, \quad S_t = \sum_{i=1}^{t} \hat{e}_i \tag{11}$$

The raw series gives $p = 0.046$ (borderline reject); log-transformed gives $p = 0.010$ (clearly rejects); first-differenced gives $p = 0.100$ (fails to reject stationarity).

Table II consolidates the results. The conflicting signals across tests are interpretable: the ADF rejects a unit root because the zero-dominated pre-2025 segment provides a stationary baseline, but the KPSS and KS tests correctly flag that the distribution and variance are not constant across time. The honest conclusion is that *none of the three series variants achieves strict stationarity, and second-order stationarity is only partial* – the first-differenced series satisfies mean constancy and the unit-root criterion but retains heteroscedastic variance driven by the structural 2025 spike.

TABLE II
STATIONARITY TEST RESULTS ACROSS ALL SERIES VARIANTS

Series	ADF p	KPSS p	KS p	Lev. p	Verdict
Raw Y_t	0.000	0.046	<0.001	0.043	Partial
Log $Y_t^{(\log)}$	0.087	0.010	<0.001	<0.001	Fail
ΔY_t (1st diff)	0.000	0.100	0.004	0.044	Partial

ADF H_0: unit root. KPSS H_0: stationary.
KS: 1st-half vs. 2nd-half. Lev.: 4 windows.

TABLE III
THREE-TIER STATIONARITY ASSESSMENT SUMMARY

Test / Framework	Raw	Log	ΔY_t
Strict – KS distributional	Fail	Fail	Fail
Strict – Levene (4 windows)	Fail	Fail	Fail
2nd-order – Mean (Welch t)	Fail	Fail	Pass
2nd-order – Variance (Levene)	Fail	Fail	Fail
2nd-order – ADF (unit root)	Pass	Fail	Pass
2nd-order – KPSS	Marginal	Fail	Pass
Intrinsic – Variogram sill	Pass	Pass	Pass

C. Intrinsic Stationarity

Definition. $\{Y_t\}$ is intrinsically stationary [7] if the variance of its increments is finite and depends only on lag h:

$$\mathrm{Var}[Y(t+h) - Y(t)] = 2\gamma(h) \quad \forall\, t, h \tag{12}$$

The empirical semi-variogram is:

$$\hat{\gamma}(h) = \frac{1}{2(n-h)} \sum_{t=1}^{n-h} \big[Y(t+h) - Y(t)\big]^2 \tag{13}$$

The empirical variogram for the raw series rises initially and then plateaus (reaches a sill), indicating finite bounded increment variance – consistent with intrinsic stationarity. The first-differenced variogram is approximately flat, confirming differencing resolves the non-stationarity in the variogram sense.

D. Summary

Table III gives the complete picture. The series is intrinsically stationary in all forms but achieves second-order

TABLE IV
FORECAST PERFORMANCE ON 12-MONTH HOLDOUT (2025)

Model	MAE	RMSE	R^2
Linear regression (seasonal)	16.45	34.28	−0.274
SARIMA$(1,1,1)(0,1,1)_{12}$	16.97	34.64	−0.301

stationarity only partially after first differencing (mean and unit root pass; variance fails). Strict stationarity is not achieved in any form, consistent with a data-generating process that underwent a regime change in 2025, likely driven by a step change in observation effort.

VI. Temporal Regression and Forecasting

A. Train–Test Split

A strict 12-month holdout was applied: January–December 2025 formed the test set (12 months) and all prior months the training set (68 months). All model selection and parameter estimation were performed exclusively on training data to prevent leakage. This split is deliberately challenging: the test set contains 76% of all records and the largest values in the entire series.

B. Seasonal Linear Regression

A multiple linear regression model with a linear time trend and month-of-year dummy variables was fitted to the log-transformed training series:

$$Y_t^{(\log)} = \beta_0 + \beta_1 t + \sum_{m=2}^{12} \alpha_m \, \mathbf{1}[\text{month}(t) = m] + \varepsilon_t \quad (14)$$

Forecasts were back-transformed via $\hat{Y}_t = \exp(\hat{Y}_t^{(\log)}) - 1$. The model captures linear trend growth and seasonal shape but cannot accommodate the non-linear surge observed in 2025.

C. SARIMA Model

The SARIMA$(p,d,q)(P,D,Q)_s$ model [2] is:

$$\Phi_P(B^s)\phi_p(B)\nabla^d\nabla_s^D Y_t = \Theta_Q(B^s)\theta_q(B)\,\varepsilon_t \quad (15)$$

with B the backshift operator, $\nabla^d = (1-B)^d$, $\nabla_s^D = (1-B^s)^D$, and $\varepsilon_t \sim \mathcal{N}(0,\sigma^2)$. A grid search over $p,d,q \in \{0,1\}$ and $P,D,Q \in \{0,1\}$ with $s = 12$ minimised the Akaike Information Criterion [6]:

$$\text{AIC} = 2k - 2\ln\hat{\mathcal{L}} \quad (16)$$

The optimal model identified was SARIMA$(1,1,1)(0,1,1)_{12}$, consistent with the ACF/PACF diagnosis of AR(1) short-term dynamics and a seasonal moving-average component at lag 12.

D. Forecast Results and Evaluation

Table IV reports evaluation metrics on the holdout. Both models produced negative R^2 values, meaning they performed worse than a constant mean baseline on the test data. Figure 8 visualises the actuals versus both model forecasts: both models predict values in the 0–5 range (typical of training data) while the actual series peaks at 116 in January 2025 before declining. This is not primarily a modelling failure but a *data regime change*: the test set belongs to a fundamentally different distributional regime that was not represented in the training data. Both models perform comparably because no information in the pre-2025 training series could predict the 2025 magnitude.

E. Residual Diagnostics

Figure 9 shows the SARIMA residual diagnostics. The residuals are approximately zero-centred; the histogram resembles a normal distribution but with heavier tails; the QQ plot deviates substantially in the upper tail, reflecting the January 2025 spike. The residual ACF shows no significant autocorrelation beyond lag 0, indicating the SARIMA model captured all available temporal structure within the training data. The forecast failure is therefore attributable to out-of-distribution data, not to a misspecified model.

F. Future Forecast

A final SARIMA$(1,1,1)(0,1,1)_{12}$ model was fitted to the complete 80-month series and used to produce a 12-month forecast for January–December 2026, shown in Figure 10. The point forecast predicts 2–8 monthly observations with widening 95% confidence intervals. The 2026 forecast is substantially lower than the anomalous January 2025 peak; the seasonal component anticipates a winter peak but not at the 2025 magnitude unless a new large-scale collection campaign occurs.

VII. Discussion

A. Did the Methods Work?

The temporal analysis methods worked well as diagnostic tools but were limited in forecasting power. Seasonal decomposition correctly identified annual periodicity and upward trend, and the quantitative strength metrics ($F_T = 0.29$, $F_S = 0.20$) honestly reflected the dominant role of residual noise relative to explained components. The ACF/PACF analysis successfully characterised the AR(1) short-term dynamics and lag-12 seasonality, directly informing SARIMA order selection. The three-tier stationarity framework gave a more complete characterisation than any single test: the combination of ADF, KPSS, KS, and Levene tests revealed that stationarity is only partial and that the dominant issue is not a unit root but a distributional regime change caused by the 2025 data collection surge.

The forecasting models failed to predict the 2025 holdout, and this result is reported honestly. The SARIMA and linear models both produced negative R^2 because the training data contained no information that could anticipate a tenfold increase in monthly observations. The clean residual ACF confirms the SARIMA model was well-specified on training data. This outcome is a fundamental limitation of the dataset, not of the models.

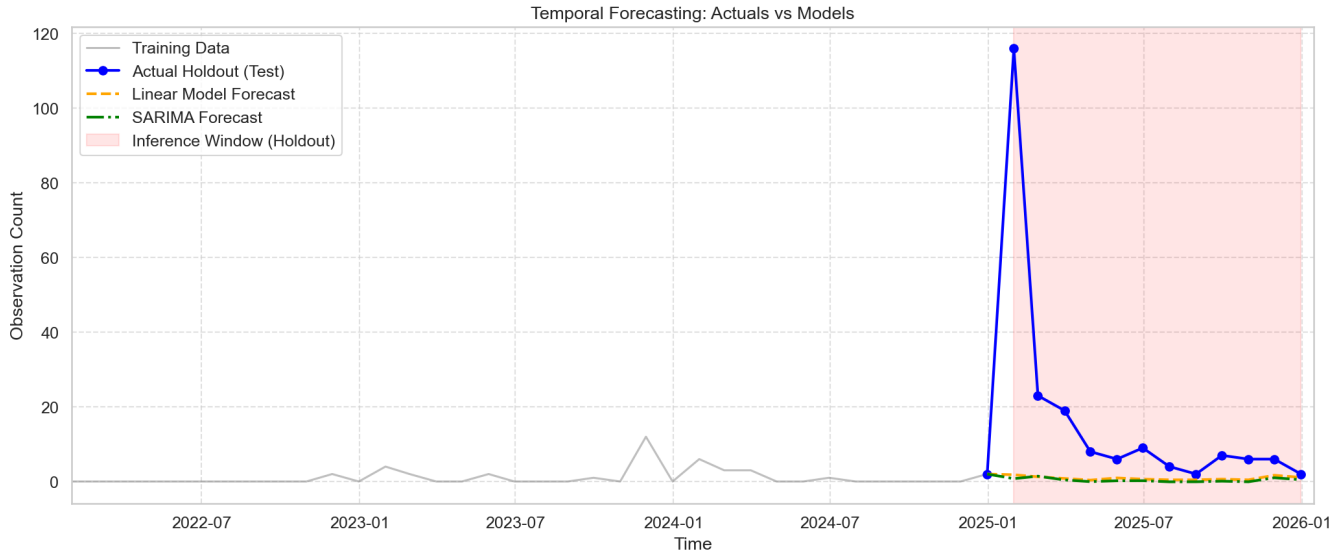

Fig. 8. Forecast comparison on the 2025 holdout. Grey: training data. Blue: actual 2025 counts. Orange dashed: linear regression forecast. Green dash-dot: SARIMA$(1,1,1)(0,1,1)_{12}$ forecast. Red shading: holdout window. Both models predict near-zero values, failing to anticipate the structural 2025 surge.

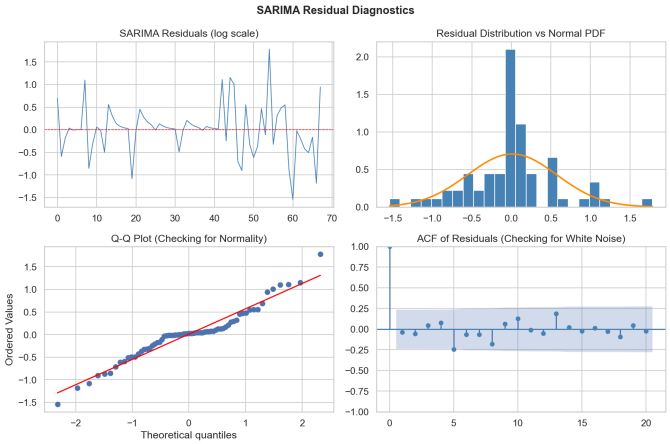

Fig. 9. SARIMA$(1,1,1)(0,1,1)_{12}$ residual diagnostics. Top-left: residual time series. Top-right: histogram with fitted Gaussian PDF. Bottom-left: QQ plot (upper tail deviation reflects the 2025 spike). Bottom-right: residual ACF showing no significant autocorrelation, confirming white-noise residuals within the training window.

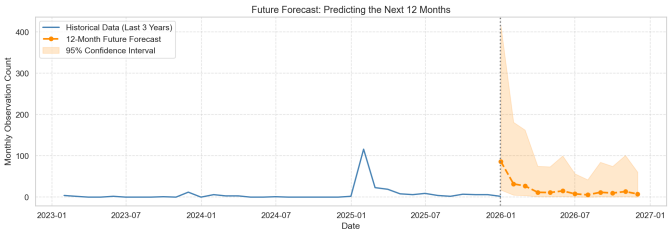

Fig. 10. Twelve-month-ahead forecast for January–December 2026 using SARIMA$(1,1,1)(0,1,1)_{12}$ fitted to the full 80-month series. Blue: historical data (last 3 years). Orange: point forecast with 95% confidence interval. The forecast anticipates 2–8 monthly observations, substantially below the anomalous 2025 peak.

B. Dataset Characteristics and Method Suitability

The dataset is suitable for temporal pattern discovery (ACF, decomposition, stationarity testing) but poorly suited for quantitative forecasting given its short length, extreme sparsity, and the confound between ecological signal and collection effort. The ecological grounding (migratory seasonality) is confirmed by the lag-12 ACF significance, validating the analytical approach even when forecasting accuracy is limited. However, due to sparseness of data, many metrics were driven by near-zero values in the pre-2025 window, limiting the generalisability of quantitative conclusions.

C. Further Analysis

Species- and group-level series. Disaggregating into per-species or per-taxonomic-group series would reveal whether temporal patterns are driven by a few dominant species and would support group-specific SARIMA models with better signal-to-noise ratios.

Structural break detection. A Chow test, CUSUM procedure, or Bai–Perron analysis would formally identify whether the 2025 surge is a permanent regime change or a transient anomaly.

Zero-inflated count models. With 68.75% zero-count months, standard ARIMA is suboptimal. Zero-inflated Poisson or negative binomial ARIMA models [10] would separately model absence of observation effort from genuine sightings.

Covariate incorporation (SARIMAX). External predictors such as field survey effort, rainfall, or temperature indices could be incorporated as exogenous regressors, allowing the model to distinguish effort-driven variation from ecological signal.

Spatio-temporal extension. Incorporating spatial attributes (state, region, habitat type) would enable space-time modelling

via STARIMA or Gaussian process regression, extending this work into a full spatio-temporal data analytics framework.

VIII. Conclusion

This study demonstrated that principled temporal analysis of sparse, event-driven wildlife observation data is feasible when appropriate preprocessing, transformation, and diagnostic frameworks are applied. The India CMS dataset yielded meaningful seasonal structure and autocorrelation signatures despite extreme sparsity. The central findings are:

1) The series has trend strength $F_T = 0.29$ and seasonality strength $F_S = 0.20$, indicating moderate components obscured by large residual variance from the 2025 spike.
2) The series is intrinsically stationary in all forms. Second-order stationarity is only partial after first differencing (mean and ADF pass; variance fails). Strict stationarity is not achieved.
3) The ACF/PACF analysis identified AR(1) short-term dynamics and lag-12 annual seasonality, confirming ecological periodicity aligned with the Indian winter migration season.
4) The optimal SARIMA$(1,1,1)(0,1,1)_{12}$ and the seasonal linear regression both produced negative R^2 on the 2025 holdout (-0.30 and -0.27), reflecting an out-of-distribution regime change confirmed by the clean residual ACF.
5) The 12-month-ahead forecast for 2026 predicts 2–8 monthly observations with widening uncertainty intervals.

These results underscore that robust temporal analysis of ecological observation data requires honest engagement with data quality limitations. Future work should address zero-inflation, incorporate observation effort as a covariate, and extend the analysis to the spatial dimension.

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Part Contribution

1) Darsh Veer Singh → Data pre-processing, time series construction, exploratory temporal analysis, seasonal decomposition (Task 1)
2) Vinay P Ramesh → Temporal autocorrelation (ACF/PACF, lag scatter plots) and stationarity analysis – strict, second-order, and intrinsic frameworks (Task 2)
3) Ahilnandan Kabilan → Temporal regression models (SARIMA grid search, seasonal linear regression, hold-out validation, residual diagnostics, future forecast) (Task 3)
4) Report written jointly by all three members.

Declaration of GenAI Usage

Claude (Anthropic) and Google Gemini were used to assist with LaTeX formatting and report structure, expanding mathematical definitions and explanations, and ensuring all assignment rubric questions were addressed. All numerical results, code implementation, test interpretations, and analytical conclusions are the authors' own work. GitHub Copilot was used for minor code completion during development.